

Supplementary for Counting the Unseen: Non-Contact Fruit Counting and Volume-Density Estimation Under Occlusion

I. METHODOLOGY

Our methodology consists of two closely coupled components: (i) a novel IoT-based hardware and firmware platform for acquiring structured Channel State Information (CSI), and (ii) a multimodal learning model that fuses CSI with RGBD geometry for physical reasoning in occluded object piles. This section first describes the proposed hardware and firmware design.

A. Part I: IoT-Based CSI Hardware Design

Motivation and Design Objective

Commodity ESP32 devices expose CSI measurements through the Wi-Fi receiver, but they are primarily designed for communication and networking rather than object-level perturbation sensing. Existing CSI interfaces generally provide per-packet, subcarrier-dependent channel coefficients to support wireless sensing, localization, or activity recognition [1], [2]. They do not, by themselves, identify whether a particular multipath component interacted with the object under investigation or whether it originated from a static environmental reflector.

Our objective is therefore to redesign the ESP32-based sensing platform at both the hardware and firmware levels. The hardware increases spatial diversity through an antenna array, while the firmware augments each CSI observation with sender, timing, subcarrier, and directional metadata. The resulting system is intended to transform CSI from a communication-side diagnostic signal into a structured physical sensing medium for detecting object-induced propagation perturbations.

ESP32 and External Antenna Interface

We use ESP32 IoT devices because they provide a low-cost, programmable Wi-Fi platform with access to packet-level CSI measurements. The CSI reported by an ESP32 contains complex-valued channel coefficients over OFDM subcarriers, thereby exposing amplitude and phase changes caused by propagation through the environment [1]. To improve spatial selectivity and link budget, we replace the onboard antenna interface with an externally soldered 6-dBi antenna.

The external antenna serves three purposes. First, its higher gain improves the received signal-to-noise ratio (SNR), which is important because object-induced multipath perturbations may be small relative to the direct and static background components. Second, a known external

antenna provides a more reproducible radiation and reception pattern than an unconstrained onboard placement. Third, the external connector allows the antenna geometry and orientation to be calibrated and repeated across the array. The antenna gain does not create directional CSI by itself; rather, it improves link reliability and makes the spatial arrangement of the antenna elements more controllable.

For transmitter m and receiver n , the calibrated antenna locations are denoted by

$$\mathbf{p}_m^{\text{tx}} \in \mathbb{R}^3, \quad \mathbf{p}_n^{\text{rx}} \in \mathbb{R}^3. \quad (1)$$

The corresponding antenna orientations are represented by

$$\mathbf{R}_m^{\text{tx}}, \mathbf{R}_n^{\text{rx}} \in SO(3). \quad (2)$$

3 × 3 Transmitter and Receiver Arrays

We arrange nine ESP32 transmitters in a 3×3 planar array and independently arrange nine ESP32 receivers in another 3×3 planar array. Let

$$\mathcal{M} = \{1, \dots, 9\}, \quad \mathcal{N} = \{1, \dots, 9\} \quad (3)$$

denote the transmitter and receiver indices. The array can be generalized to an $N_x \times N_y$ configuration, with

$$N_{\text{elem}} = N_x N_y. \quad (4)$$

For a regular planar array with element spacing d_x and d_y , the position of element (u, v) is

$$\mathbf{p}_{u,v} = \mathbf{p}_0 + u d_x \mathbf{e}_x + v d_y \mathbf{e}_y, \quad (5)$$

where

$$u \in \{0, \dots, N_x - 1\}, \quad v \in \{0, \dots, N_y - 1\}. \quad (6)$$

The transmitter and receiver arrays produce a set of 81 nominal transmitter–receiver links:

$$\mathcal{L} = \{(m, n) : m \in \mathcal{M}, n \in \mathcal{N}\}, \quad |\mathcal{L}| = 81. \quad (7)$$

The aggregate set of links provides spatial diversity across the pile. A direct path, a path reflected by the pile, and a path passing through a partially transmitting object generally produce different delay, phase, amplitude, and direction responses across the 81 links. The array therefore provides a spatially distributed sampling field rather than relying on one transmitter–receiver measurement.

The term “dense CSI field” in this work refers to the spatially distributed collection

$$\mathcal{H} = \{H_{mn,s}(t) : (m, n) \in \mathcal{L}, s \in \mathcal{S}, t \in \mathcal{T}\}, \quad (8)$$

rather than a literal continuous electromagnetic field or an X-ray image. The field becomes denser as the number of array elements, subcarriers, and temporal samples increases.

Physical CSI Observation Model

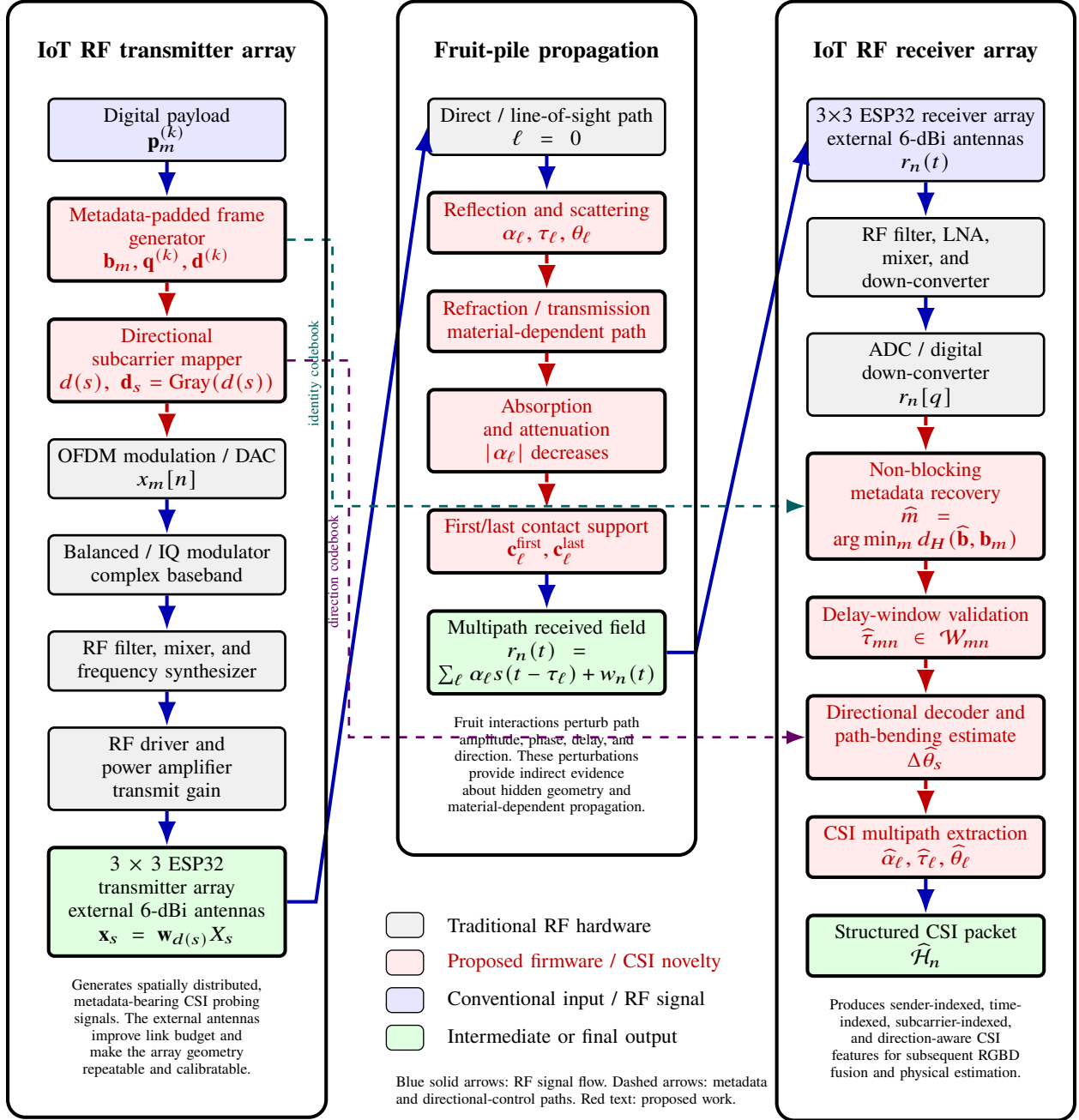


Fig. 1: Workflow of the proposed hardware interfacing for CSI array and firmware design for structured CSI acquisition.

For transmitter m , receiver n , subcarrier s , and time index t , the measured CSI coefficient is modeled as

$$H_{mn,s}(t) = \sum_{\ell=0}^{L_{mn}(t)} \alpha_{mn,\ell}(t) \exp(-j2\pi f_s \tau_{mn,\ell}(t)) + \varepsilon_{mn,s}(t), \quad (9)$$

where $\ell = 0$ denotes a direct or reference path and $\ell \geq 1$ denotes object- or environment-induced multipath components. Here,

$$\alpha_{mn,\ell}(t) = a_{mn,\ell}(t) \exp(j\varphi_{mn,\ell}(t)) \quad (10)$$

is the complex path gain, $\tau_{mn,\ell}(t)$ is the propagation delay,

f_s is the s -th subcarrier frequency, and $\varepsilon_{mn,s}(t)$ represents receiver noise, quantization error, and unmodeled interference.

The magnitude and phase of the CSI are

$$A_{mn,s}(t) = |H_{mn,s}(t)|, \quad (11)$$

$$\Phi_{mn,s}(t) = \arg(H_{mn,s}(t)). \quad (12)$$

If the CSI is stored as in-phase and quadrature components,

$$H_{mn,s}(t) = I_{mn,s}(t) + jQ_{mn,s}(t), \quad (13)$$

then

$$A_{mn,s}(t) = \sqrt{I_{mn,s}^2(t) + Q_{mn,s}^2(t)}, \quad (14)$$

$$\Phi_{mn,s}(t) = \text{atan2}(Q_{mn,s}(t), I_{mn,s}(t)). \quad (15)$$

The subcarrier frequency is

$$f_s = f_c + (s - s_0)\Delta f, \quad (16)$$

where f_c is the center frequency, Δf is the OFDM subcarrier spacing, and s_0 is the center-subcarrier index. The phase associated with path ℓ is approximately

$$\Phi_{mn,s}^{(\ell)}(t) = \varphi_{mn,\ell}(t) - 2\pi f_s \tau_{mn,\ell}(t). \quad (17)$$

Object-Induced Multipath Perturbations

When the transmitted wave interacts with the object pile, its propagation may be divided into reflected, scattered, refracted, transmitted, and absorbed components. We represent each path ℓ by

$$\pi_{mn,\ell} = (\alpha_{mn,\ell}, \tau_{mn,\ell}, \theta_{mn,\ell}^{\text{tx}}, \theta_{mn,\ell}^{\text{rx}}, \chi_{mn,\ell}), \quad (18)$$

where θ^{tx} and θ^{rx} are departure and arrival directions and $\chi_{mn,\ell}$ is a path-interaction label or latent path type.

The path length is

$$d_{mn,\ell} = \|\Gamma_{mn,\ell}\|, \quad (19)$$

where $\Gamma_{mn,\ell}$ denotes the complete propagation trajectory. Its delay is

$$\tau_{mn,\ell} = \frac{d_{mn,\ell}}{v}, \quad (20)$$

where v is the propagation velocity.

A path that undergoes additional reflections generally has a longer path length and therefore a larger delay. Its received amplitude is also reduced by the product of path-dependent attenuation factors:

$$a_{mn,\ell} = a_{mn,0} \prod_{r=1}^{R_{mn,\ell}} \gamma_{mn,\ell,r}, \quad 0 < \gamma_{mn,\ell,r} \leq 1, \quad (21)$$

where $R_{mn,\ell}$ is the number of interactions and $\gamma_{mn,\ell,r}$ represents reflection, scattering, transmission, or absorption loss at interaction r . Thus, more complex multipath trajectories commonly yield lower amplitudes, although constructive and destructive interference can cause the total measured CSI amplitude to vary non-monotonically.

Relative to a reference scene without the object pile, the object-induced CSI perturbation is

$$\Delta A_{mn,s}(t) = A_{mn,s}^{\text{pile}}(t) - A_{mn,s}^{\text{ref}}(t), \quad (22)$$

$$\Delta \Phi_{mn,s}(t) = \text{wrap} \left[\Phi_{mn,s}^{\text{pile}}(t) - \Phi_{mn,s}^{\text{ref}}(t) \right], \quad (23)$$

$$\Delta \tau_{mn,\ell}(t) = \tau_{mn,\ell}^{\text{pile}}(t) - \tau_{mn,\ell}^{\text{ref}}(t). \quad (24)$$

These perturbations are the physical measurements used by the proposed firmware. They do not directly expose a complete hidden geometry. Instead, they provide indirect evidence about object position, surface interaction, material-dependent attenuation, and the geometry of paths that intersect or pass near the pile.

Firmware Extension I: Metadata-Padded CSI Frames

The standard CSI callback provides a channel estimate associated with a received packet, but it does not necessarily provide sufficient application-level information to iden-

tify the transmitting array element, distinguish delayed multipath observations, or recover a desired acquisition order under asynchronous operation. We therefore define an application-layer metadata structure embedded in each transmitted frame.

For transmitter m and frame index k , the proposed frame is

$$\mathbf{s}_m^{(k)} = [\mathbf{p}_m^{(k)}; \mathbf{b}_m; \mathbf{q}^{(k)}; \mathbf{d}^{(k)}], \quad (25)$$

where:

- $\mathbf{p}_m^{(k)}$ is the payload;
- $\mathbf{b}_m$ is a transmitter identity code;
- $\mathbf{q}^{(k)}$ is a frame sequence, timestamp, and error-detection field;
- $\mathbf{d}^{(k)}$ contains subcarrier direction metadata.

The transmitter identity codes are selected from a codebook

$$\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_{|\mathcal{M}|}\} \quad (26)$$

with minimum Hamming distance

$$d_{\min} = \min_{i \neq j} d_H(\mathbf{b}_i, \mathbf{b}_j). \quad (27)$$

At receiver n , the decoded metadata is $\widehat{\mathbf{b}}_n^{(r)}$. The sender is identified by

$$\widehat{m} = \arg \min_{m \in \mathcal{M}} d_H(\widehat{\mathbf{b}}_n^{(r)}, \mathbf{b}_m). \quad (28)$$

The frame is accepted if

$$d_H(\widehat{\mathbf{b}}_n^{(r)}, \mathbf{b}_{\widehat{m}}) \leq \eta, \quad 2\eta < d_{\min}. \quad (29)$$

This operation is non-blocking: the receiver does not wait for a predetermined transmitter slot. It continuously processes arrived frames, identifies their origin through metadata correlation, and places accepted CSI observations into a timestamped buffer:

$$\mathcal{Q}_n \leftarrow \mathcal{Q}_n \cup \{\widehat{m}, \widehat{k}, t_n^{(r)}, \widehat{\mathbf{H}}_n^{(r)}\}. \quad (30)$$

A delay-tolerance window is used to reject implausible associations:

$$\mathcal{W}_{mn} = [\underline{\tau}_{mn}, \overline{\tau}_{mn}], \quad (31)$$

with

$$\widehat{\tau}_{mn}^{(r)} = t_n^{(r)} - \widehat{t}_m^{(k)} - \widehat{\Delta}_{mn}. \quad (32)$$

An observation is retained if

$$\widehat{\tau}_{mn}^{(r)} \in \mathcal{W}_{mn}. \quad (33)$$

Firmware Extension II: Directional Subcarrier Metadata

To associate subcarrier observations with nominal spatial directions, we divide the angular field of interest into D clockwise direction bins:

$$\Theta = \{\theta_0, \dots, \theta_{D-1}\}, \quad \theta_d = \frac{2\pi d}{D}. \quad (34)$$

Each subcarrier s is assigned a direction index

$$d(s) = (s + s_{\text{off}}) \bmod D, \quad (35)$$

where s_{off} is a configurable offset. The direction index is encoded with a Gray code:

$$\mathbf{d}_s = \text{Gray}_{B_d}(d(s)), \quad B_d = \lceil \log_2 D \rceil. \quad (36)$$

The metadata can be carried in reserved pilot groups, an application-layer extension field, or a low-rate side channel associated with the CSI acquisition frame. The

implementation must preserve the normal OFDM waveform and must not assume that an arbitrary bit can be inserted into an already standardized PHY preamble without corresponding receiver support.

At the receiver, the directional metadata is decoded as

$$\widehat{d}_{\text{rx}}(s) = \text{Gray}^{-1}(\widehat{\mathbf{d}}_s). \quad (37)$$

The circular direction displacement is

$$\Delta d_s = \left(\widehat{d}_{\text{rx}}(s) - d(s) \right)_{\text{circ}}, \quad (38)$$

and the corresponding angular displacement is

$$\Delta \theta_s = \frac{2\pi}{D} \Delta d_s. \quad (39)$$

The direction code should be interpreted as an encoded nominal transmission or beam index. A passive propagation path does not literally rewrite the transmitted metadata bits. Therefore, the observed direction change is inferred from the decoded direction state together with receive-array spatial processing, CSI phase/amplitude behavior, and path estimation. In the proposed system, the receiver forms the directional path descriptor

$$\widehat{\pi}_{mn,s,\ell} = \left[\widehat{A}_{mn,s,\ell}, \widehat{\Phi}_{mn,s,\ell}, \widehat{\tau}_{mn,\ell}, \widehat{\theta}_{mn,\ell}^{\text{rx}}, \widehat{\Delta\theta}_{mn,s,\ell}, \widehat{m}, \widehat{k} \right]. \quad (40)$$

First-Contact and Last-Contact Representation

The proposed physical-layer representation does not claim that a single CSI coefficient uniquely reveals the exact first and last physical contact points. Instead, it estimates a probabilistic contact support from the array geometry, delay, direction, and path amplitude.

For path (m, n, ℓ) , let $\Gamma_{mn,\ell}$ be the estimated path trajectory. We define:

$$\mathbf{c}_{mn,\ell}^{\text{first}} = \Gamma_{mn,\ell} \left(u_{mn,\ell}^{\text{first}} \right), \quad (41)$$

$$\mathbf{c}_{mn,\ell}^{\text{last}} = \Gamma_{mn,\ell} \left(u_{mn,\ell}^{\text{last}} \right), \quad (42)$$

where u^{first} and u^{last} are latent normalized path coordinates satisfying

$$0 \leq u_{mn,\ell}^{\text{first}} \leq u_{mn,\ell}^{\text{last}} \leq 1. \quad (43)$$

The contact-support probability is predicted as

$$p_{mn,\ell}^{\text{contact}} = \sigma \left(g_{\text{contact}} \left(\widehat{A}_{mn,\ell}, \widehat{\Phi}_{mn,\ell}, \widehat{\tau}_{mn,\ell}, \widehat{\theta}_{mn,\ell}, \widehat{\Delta\theta}_{mn,\ell} \right) \right). \quad (44)$$

A path-weighted contact descriptor is then defined by

$$\mathbf{z}_{mn,\ell}^{\text{contact}} = \left[\mathbf{c}_{mn,\ell}^{\text{first}}, \mathbf{c}_{mn,\ell}^{\text{last}}, p_{mn,\ell}^{\text{contact}}, \widehat{A}_{mn,\ell}, \widehat{\tau}_{mn,\ell}, \widehat{\Delta\theta}_{mn,\ell} \right]. \quad (45)$$

The amplitude is normalized relative to a reference or direct path:

$$\widetilde{A}_{mn,\ell} = \frac{|\widehat{\alpha}_{mn,\ell}|}{|\widehat{\alpha}_{mn,0}| + \epsilon}. \quad (46)$$

The corresponding path complexity feature is

$$\kappa_{mn,\ell} = -\log \left(\widetilde{A}_{mn,\ell} + \epsilon \right) + \lambda_{\tau} \left(\widehat{\tau}_{mn,\ell} - \widehat{\tau}_{mn,0} \right) + \lambda_{\theta} \left| \widehat{\Delta\theta}_{mn,\ell} \right|. \quad (47)$$

Larger $\kappa_{mn,\ell}$ indicates stronger attenuation, excess delay, and/or angular deviation, but it is interpreted as evidence of path interaction rather than as a deterministic label for a particular physical mechanism.

End-to-End Hardware Data Product

After processing, each receiver produces a structured CSI observation packet:

$$\widehat{\mathcal{H}}_n = \left\{ \widehat{\mathbf{H}}_{mn}^{(k)}, \widehat{m}, \widehat{k}, \widehat{\tau}_{mn,\ell}, \widehat{A}_{mn,s,\ell}, \widehat{\Phi}_{mn,s,\ell}, \widehat{\theta}_{mn,\ell}^{\text{rx}}, \widehat{\Delta\theta}_{mn,s,\ell}, \mathbf{z}_{mn,\ell}^{\text{contact}} \right\}. \quad (48)$$

The complete array-level hardware output is

$$\widehat{\mathcal{H}} = \bigcup_{n \in \mathcal{N}} \widehat{\mathcal{H}}_n. \quad (49)$$

This structured output is subsequently fused with the RGBD point cloud in Part II of the methodology. The hardware therefore does not merely collect raw CSI amplitudes; it produces sender-indexed, time-indexed, subcarrier-indexed, direction-aware, and path-aware observations suitable for multimodal geometric inference.

Hardware and Firmware Pipeline

Figure 1 summarizes the proposed hardware and firmware design. Gray blocks denote conventional RF components, whereas red blocks identify firmware or signal-processing additions introduced for asynchronous CSI reception and directional path analysis.

$$\widehat{\mathbf{z}}_{mn,\ell} = \left[\widehat{\mathbf{c}}_{mn,\ell}^{\text{first}}, \widehat{\mathbf{c}}_{mn,\ell}^{\text{last}}, \widehat{\alpha}_{mn,\ell}, \widehat{\tau}_{mn,\ell}, \widehat{\theta}_{mn,\ell}, \widehat{\Delta\theta}_{mn,\ell} \right]. \quad (50)$$

Firmware Metadata, Static-Reflector Suppression, and CSI Sampling

The firmware augments each CSI-bearing frame with a transmitter identifier, frame counter, local timestamp, error-detection code, and directional subcarrier metadata, enabling asynchronous sender association and delay-window validation without strict global synchronization. Static reflectors are characterized during an empty-scene calibration stage and removed using temporal CSI residuals, delay/angle consistency, and spatial agreement with the RGBD pile region; perturbations that persist within the object-supported region are retained as object-related multipath evidence. The CSI sampling rate is hardware- and firmware-dependent and is reported as the measured valid-frame rate

$$R_{\text{CSI}} = \frac{N_{\text{valid}}}{T_{\text{obs}}}, \quad (51)$$

while the aggregate array rate is

$$R_{\text{agg}} = |\mathcal{L}| R_{\text{CSI}}, \quad |\mathcal{L}| = N_{\text{tx}} N_{\text{rx}}. \quad (52)$$

Accordingly, the final paper should report the experimentally measured per-link and aggregate rates rather than assuming a nominal CSI sampling frequency.

B. Part II: Multimodal Model Design

Problem Formulation and Notation

Let a scene contain a set of partially observed objects indexed by $i \in \{1, \dots, N\}$. The input to the proposed framework is a synchronized RGBD observation and a CSI sequence acquired from the same object pile. The partial RGBD point cloud is represented as

$$\mathcal{P}^{\text{RGBD}} = \{\mathbf{p}_j\}_{j=1}^J, \quad \mathbf{p}_j = [\mathbf{x}_j, \mathbf{c}_j] \in \mathbb{R}^6,$$

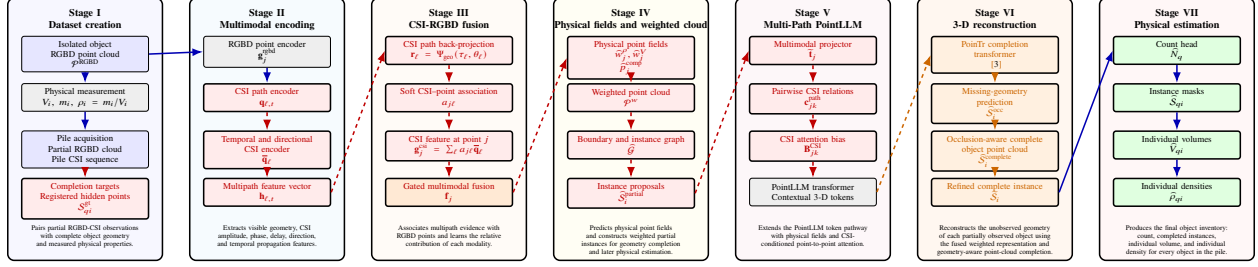


Fig. 2: Extended architecture of the proposed multimodal RGBD-CSI Multi-Path PointLLM framework. The method encodes RGBD geometry and temporal CSI multipath features, performs geometry-aware CSI-RGBD fusion, constructs weighted partial instances, and processes the fused representation using a Multi-Path PointLLM. A PoinTr-based 3-D reconstruction module then predicts the missing geometry of partially observed objects and produces occlusion-aware complete object point clouds. The completed instances are finally used to estimate object count, instance masks, individual volume, and individual density. Red blocks denote the proposed CSI-processing, multimodal-fusion, physical-field, and CSI-conditioned PointLLM components. Orange blocks denote the PoinTr-based 3-D reconstruction module. Gray blocks represent conventional or inherited processing components, blue blocks denote inputs, and green blocks indicate final physical-inventory outputs.

where $\mathbf{x}_j \in \mathbb{R}^3$ denotes the spatial coordinate of point j and $\mathbf{c}_j \in \mathbb{R}^3$ denotes its RGB color. The synchronized CSI sequence is denoted by

$$\mathcal{H} = \{\mathbf{H}_{n,t}\}_{n=1, t=1}^{N_{rx}, T}, \quad \mathbf{H}_{n,t} = [H_{n,t,1}, \dots, H_{n,t,K}],$$

where n indexes the receiver channel, t indexes time, and K is the number of CSI subcarriers. Each complex CSI coefficient is decomposed as

$$H_{n,t,k} = A_{n,t,k} \exp(i\phi_{n,t,k}),$$

where $A_{n,t,k}$ and $\phi_{n,t,k}$ are the calibrated amplitude and phase, respectively. The objective is to learn a mapping

$$\mathcal{F}_\Theta : (\mathcal{P}^{\text{RGBD}}, \mathcal{H}) \rightarrow (\hat{N}, \hat{\mathcal{S}}_i^{\text{complete}}, \hat{V}_i, \hat{\rho}_i),$$

where $\hat{N}$ is the estimated object count, $\hat{\mathcal{S}}_i^{\text{complete}}$ is the reconstructed complete point cloud of object i , and $\hat{V}_i$ and $\hat{\rho}_i$ are its estimated volume and density.

Stage I: RGBD-CSI Data Acquisition and Preparation

The first stage produces calibrated and synchronized inputs for the remaining modules. For each RGBD point $\mathbf{p}_j$, the spatial coordinate $\mathbf{x}_j$ and color vector $\mathbf{c}_j$ are obtained from the calibrated RGBD sensor. The raw CSI coefficient $H_{n,t,k}$ is converted into amplitude and phase using

$$A_{n,t,k} = |H_{n,t,k}|, \quad \phi_{n,t,k} = \arg(H_{n,t,k}).$$

After phase unwrapping, amplitude normalization, and temporal synchronization, the CSI input is represented by

$$\mathbf{z}_{n,t,k} = [\tilde{A}_{n,t,k}, \tilde{\phi}_{n,t,k}, \Delta\tilde{\phi}_{n,t,k}],$$

where $\tilde{A}_{n,t,k}$ and $\tilde{\phi}_{n,t,k}$ are normalized amplitude and phase values, and $\Delta\tilde{\phi}_{n,t,k}$ is the temporal phase difference. The prepared inputs to the next stage are therefore the RGBD point set $\mathcal{P}^{\text{RGBD}}$ and the calibrated CSI tensor

$$\mathcal{Z} = \{\mathbf{z}_{n,t,k}\}_{n,t,k}.$$

Stage II: RGBD and Temporal CSI Encoding

The RGBD encoder maps each point $\mathbf{p}_j$ to a geometric feature:

$$\mathbf{g}_j^{\text{rgb}} = E_{\text{rgb}}(\mathbf{p}_j) = E_{\text{rgb}}([\mathbf{x}_j, \mathbf{c}_j]) \in \mathbb{R}^{D_g}.$$

The complete RGBD feature set is

$$\mathcal{G}^{\text{rgb}} = \{\mathbf{g}_j^{\text{rgb}}\}_{j=1}^J.$$

In parallel, the normalized CSI sequence $\mathcal{Z}$ is processed by a temporal CSI encoder. For a CSI path indexed by ℓ , the encoder receives the corresponding sequence

$$\mathbf{Z}_\ell = \{\mathbf{z}_{\ell,t}\}_{t=1}^T$$

and produces a path-level representation

$$\bar{\mathbf{q}}_\ell = E_{\text{CSI}}(\mathbf{Z}_\ell) \in \mathbb{R}^{D_q}.$$

The temporal CSI representation is augmented with the estimated path delay τ_ℓ and direction θ_ℓ :

$$\mathbf{q}_\ell = [\bar{\mathbf{q}}_\ell, \tau_\ell, \theta_\ell].$$

Thus, the output of the encoding stage consists of the RGBD features $\mathcal{G}^{\text{rgb}}$ and the set of structured CSI path descriptors

$$\mathcal{Q} = \{\mathbf{q}_\ell\}_{\ell=1}^L,$$

where L is the number of detected multipath components.

Stage III: CSI Back-Projection and Point Association

The CSI path descriptors $\mathbf{q}_\ell$ are converted into approximate three-dimensional path locations using the estimated delay and direction. We define the geometry-aware back-projection function as

$$\mathbf{r}_\ell = \Psi_{\text{geo}}(\tau_\ell, \theta_\ell; \mathbf{K}, \mathbf{T}) \in \mathbb{R}^3,$$

where $\mathbf{K}$ denotes the calibrated camera or sensor intrinsics and $\mathbf{T}$ denotes the transformation between the CSI and RGBD coordinate systems. The resulting set of CSI path locations is

$$\mathcal{R} = \{\mathbf{r}_\ell\}_{\ell=1}^L.$$

For each RGBD point $\mathbf{x}_j$ and CSI path location $\mathbf{r}_\ell$, a soft association score is computed as

$$e_{j\ell} = -\frac{\|\mathbf{x}_j - \mathbf{r}_\ell\|_2^2}{2\sigma_r^2} + \eta \text{sim}(\mathbf{g}_j^{\text{rgb}}, \bar{\mathbf{q}}_\ell),$$

where σ_r controls the spatial association bandwidth, η controls the contribution of feature similarity, and $\text{sim}(\cdot, \cdot)$ is a normalized feature-similarity function. The normalized CSI-point association weight is

$$a_{j\ell} = \frac{\exp(e_{j\ell})}{\sum_{u=1}^L \exp(e_{ju})}.$$

Therefore, $a_{j\ell}$ quantifies the contribution of CSI path ℓ to RGBD point j . The association matrix

$$\mathbf{A} = [a_{j\ell}] \in \mathbb{R}^{J \times L}$$

is passed to the multimodal fusion stage.

Stage IV: RGBD-CSI Multimodal Fusion and Weighted Point Cloud

The point-wise CSI feature for RGBD point j is obtained by aggregating the associated CSI path descriptors:

$$\mathbf{g}_j^{\text{csi}} = \sum_{\ell=1}^L a_{j\ell} \bar{\mathbf{q}}_\ell \in \mathbb{R}^{D_q}.$$

The RGBD feature and CSI feature are then concatenated and projected into a common feature space:

$$\mathbf{u}_j = \left[\mathbf{g}_j^{\text{rgbd}} \parallel \mathbf{g}_j^{\text{csi}} \right], \quad \mathbf{h}_j = \varphi_{\text{fuse}}(\mathbf{u}_j) \in \mathbb{R}^{D_f},$$

where $\parallel$ denotes concatenation and φ_{fuse} is a learnable multilayer fusion function. A gated fusion mechanism determines the relative contribution of the two modalities:

$$\gamma_j = \sigma(\mathbf{W}_\gamma \mathbf{u}_j + \mathbf{b}_\gamma),$$

where $\sigma(\cdot)$ is the sigmoid function, $\mathbf{W}_\gamma$ is a learnable matrix, and $\mathbf{b}_\gamma$ is a learnable bias. The unified point feature is

$$\mathbf{f}_j = \gamma_j \odot \mathbf{g}_j^{\text{rgbd}} + (\mathbf{1} - \gamma_j) \odot \mathbf{g}_j^{\text{csi}},$$

where $\odot$ denotes element-wise multiplication. If the two modality features have different dimensions, they are first projected using learned linear maps into the common dimension D_f .

The fused feature $\mathbf{f}_j$ is used to predict point-wise physical fields:

$$(\hat{w}_j^\rho, \hat{w}_j^V, \hat{p}_j^{\text{comp}}) = \Psi_{\text{field}}(\mathbf{f}_j),$$

where $\hat{w}_j^\rho$ is a learned mass- or density-related weight, $\hat{w}_j^V$ is a volume-related weight, and $\hat{p}_j^{\text{comp}} \in [0, 1]$ is the probability that point j belongs to a reconstructed or completed region. The weighted point cloud is then defined as

$$\mathcal{P}^w = \left\{ \left(\mathbf{x}_j, \mathbf{f}_j, \hat{w}_j^\rho, \hat{w}_j^V, \hat{p}_j^{\text{comp}} \right) \right\}_{j=1}^J.$$

The field predictions are also used to construct an instance graph. For two points j and u , the graph affinity is

$$G_{ju} = \exp \left(-\frac{\|\mathbf{x}_j - \mathbf{x}_u\|_2^2}{2\sigma_x^2} \right) \exp \left(-\frac{|\hat{w}_j^V - \hat{w}_u^V| + |\hat{w}_j^\rho - \hat{w}_u^\rho|}{\sigma_w} \right),$$

where σ_x and σ_w control spatial and field similarity. The resulting graph

$$\hat{\mathcal{G}} = (\mathcal{V}, \mathcal{E}, \mathbf{G})$$

contains point vertices $\mathcal{V}$, edges $\mathcal{E}$, and affinity matrix $\mathbf{G} = [G_{ju}]$. Graph partitioning produces partial instance proposals

$$\hat{\mathcal{S}}_i^{\text{partial}} \subset \mathcal{P}^w.$$

These weighted partial instances are passed to the Multi-Path PointLLM module.

Stage V: Multi-Path PointLLM Reasoning

For every point in a partial instance, the fused feature $\mathbf{f}_j$ is mapped into the PointLLM token space using a multimodal projector:

$$\tilde{\mathbf{t}}_j = \Pi_{\text{mm}}(\mathbf{f}_j) \in \mathbb{R}^{D_t},$$

where Π_{mm} is a learnable projection function and D_t is the PointLLM token dimension. The point-token sequence for

instance proposal i is

$$\tilde{\mathbf{T}}_i = \left\{ \tilde{\mathbf{t}}_j \mid \mathbf{p}_j \in \hat{\mathcal{S}}_i^{\text{partial}} \right\}.$$

To preserve multipath relationships between points, pairwise CSI relations are computed from the CSI-point association matrix:

$$\mathbf{c}_{ju}^{\text{path}} = \sum_{\ell=1}^L a_{j\ell} a_{u\ell} \mathbf{q}_\ell.$$

The relation vector is converted into an attention bias:

$$B_{ju}^{\text{CSI}} = \mathbf{w}_B^\top \mathbf{c}_{ju}^{\text{path}} + b_B,$$

where $\mathbf{w}_B$ and b_B are learnable parameters. For transformer layer r , the CSI-conditioned attention is

$$\text{Attn}_{ju}^{(r)} = \text{softmax}_u \left(\frac{\mathbf{Q}_j^{(r)} \left(\mathbf{K}_u^{(r)} \right)^\top}{\sqrt{d_k}} + B_{ju}^{\text{CSI}} \right) \mathbf{V}_u^{(r)},$$

where $\mathbf{Q}_j^{(r)}$, $\mathbf{K}_u^{(r)}$, and $\mathbf{V}_u^{(r)}$ are the query, key, and value projections at layer r , and d_k is the key dimension. The PointLLM transformer processes the projected point tokens and CSI-conditioned attention relations:

$$\mathbf{Y}_i = T_{\text{PointLLM}}(\tilde{\mathbf{T}}_i, \mathbf{B}_i^{\text{CSI}}, \Theta_{\text{pre}}),$$

where Θ_{pre} denotes the pretrained PointLLM parameters and $\mathbf{B}_i^{\text{CSI}}$ is the pairwise bias matrix for instance i . The contextual representation $\mathbf{Y}_i$ is the input to the three-dimensional reconstruction module.

Stage VI: PoinTr-Based 3D Reconstruction

The proposed framework does not treat the output of PointLLM as the final complete geometry. Instead, the contextual representation $\mathbf{Y}_i$ is provided to an explicit PoinTr-based point-cloud reconstruction module. PoinTr represents a point cloud as a sequence of unordered point groups and uses a geometry-aware transformer encoder-decoder to generate missing geometry [3]. In our framework, the contextual PointLLM representation is first converted into reconstruction tokens:

$$\mathbf{Z}_i^{\text{rec}} = \Phi_{\text{rec}}(\mathbf{Y}_i, \hat{\mathcal{S}}_i^{\text{partial}}),$$

where Φ_{rec} is a learnable reconstruction-input projection. The PoinTr encoder produces geometry-aware latent tokens:

$$\mathbf{E}_i^{\text{rec}} = E_{\text{PoinTr}}(\mathbf{Z}_i^{\text{rec}}).$$

A geometry-aware decoder then predicts point proxies for the missing region:

$$\hat{\mathcal{U}}_i = D_{\text{PoinTr}}(\mathbf{E}_i^{\text{rec}}, \hat{\mathcal{S}}_i^{\text{partial}}),$$

where $\hat{\mathcal{U}}_i$ denotes the inferred occluded point set. The occlusion-aware complete point cloud is formed by combining the observed instance points with the inferred points:

$$\hat{\mathcal{S}}_i^{\text{complete}} = \hat{\mathcal{S}}_i^{\text{partial}} \cup \hat{\mathcal{U}}_i.$$

A refinement function uses the point-wise physical fields to produce the final completed instance:

$$\tilde{\mathcal{S}}_i = R_{\text{phys}}(\hat{\mathcal{S}}_i^{\text{complete}}, \hat{w}^\rho, \hat{w}^V, \hat{p}^{\text{comp}}),$$

where R_{phys} removes geometrically inconsistent points and retains points whose predicted fields are compatible with the instance. The output $\tilde{\mathcal{S}}_i$ is passed to the physical-

estimation stage.

Stage VII: Object Count, Volume, and Density Estimation

The reconstructed instance representations are first aggregated to estimate the number of objects in the pile. Let $\mathbf{y}_i$ denote the pooled representation of reconstructed instance $\tilde{\mathcal{S}}_i$:

$$\mathbf{y}_i = \text{Pool} \left(\left\{ \mathbf{f}_j \mid \mathbf{p}_j \in \tilde{\mathcal{S}}_i \right\} \right).$$

The object-count head receives the global set of instance representations and predicts

$$\hat{N}_q = H_{\text{count}}(\mathbf{y}_1, \dots, \mathbf{y}_M),$$

where M is the number of instance proposals and $\hat{N}_q$ is the estimated number of objects in pile q .

The instance mask head assigns each reconstructed point to an object instance. For point $\mathbf{x}_j$ and instance i , the mask probability is

$$\hat{s}_{qi,j} = \sigma(H_{\text{mask}}(\mathbf{f}_j, \mathbf{y}_i)),$$

where $\hat{s}_{qi,j}$ is the probability that point j belongs to instance i in pile q . The estimated instance point set is

$$\hat{\mathcal{S}}_{qi} = \{\mathbf{x}_j : \hat{s}_{qi,j} > \delta_s\},$$

where δ_s is the mask threshold.

The individual volume is estimated by integrating the learned local volume weights over the reconstructed instance:

$$\hat{V}_{qi} = \sum_{\mathbf{x}_j \in \hat{\mathcal{S}}_{qi}} \hat{w}_j^V \Delta v_j,$$

where Δv_j is the local voxel or point-support volume associated with point j . The quantity $\hat{w}_j^V$ is the learned volume contribution predicted from the fused point feature $\mathbf{f}_j$.

Finally, the mass-related estimate for instance i is computed from the learned density or mass weight:

$$\hat{M}_{qi} = \sum_{\mathbf{x}_j \in \hat{\mathcal{S}}_{qi}} \hat{w}_j^\rho \Delta v_j.$$

The individual density is then obtained from the estimated mass and volume:

$$\hat{\rho}_{qi} = \frac{\hat{M}_{qi}}{\hat{V}_{qi} + \varepsilon},$$

where $\varepsilon > 0$ is a small numerical constant that prevents division by zero. Thus, the final output of the framework is

$$\{\hat{N}_q, \hat{\mathcal{S}}_{qi}, \hat{V}_{qi}, \hat{\rho}_{qi}\}_{i=1}^{\hat{N}_q}.$$

Training Objective:

The model is trained using a multi-task objective that supervises multimodal fusion, point-cloud completion, instance separation, counting, volume estimation, and density estimation:

$$\mathcal{L} = \lambda_{\text{comp}} \mathcal{L}_{\text{comp}} + \lambda_{\text{mask}} \mathcal{L}_{\text{mask}} + \lambda_{\text{count}} \mathcal{L}_{\text{count}} + \lambda_V \mathcal{L}_V + \lambda_\rho \mathcal{L}_\rho + \lambda_{\text{field}} \mathcal{L}_{\text{field}}. \quad (53)$$

where the λ terms control the relative contribution of each task.

The completion loss compares the reconstructed point cloud with the complete ground-truth point cloud using a

symmetric Chamfer distance:

$$\mathcal{L}_{\text{comp}} = d_{\text{CD}}(\hat{\mathcal{S}}_i^{\text{complete}}, \mathcal{S}_i^{\text{gt}}).$$

The instance-mask loss is a point-wise binary cross-entropy:

$$\mathcal{L}_{\text{mask}} = - \sum_j \left[s_{qi,j}^{\text{gt}} \log \hat{s}_{qi,j} + (1 - s_{qi,j}^{\text{gt}}) \log (1 - \hat{s}_{qi,j}) \right].$$

The count, volume, and density losses are defined as

$$\mathcal{L}_{\text{count}} = |\hat{N}_q - N_q|,$$

$$\mathcal{L}_V = \frac{1}{N_q} \sum_{i=1}^{N_q} |\hat{V}_{qi} - V_{qi}|,$$

and

$$\mathcal{L}_\rho = \frac{1}{N_q} \sum_{i=1}^{N_q} |\hat{\rho}_{qi} - \rho_{qi}|.$$

The physical-field loss supervises the predicted local volume and mass-related weights:

$$\mathcal{L}_{\text{field}} = \frac{1}{J} \sum_{j=1}^J \left[|\hat{w}_j^V - w_j^{V,\text{gt}}| + |\hat{w}_j^\rho - w_j^{\rho,\text{gt}}| + \text{BCE}(\hat{p}_j^{\text{comp}}, p_j^{\text{comp,gt}}) \right].$$

All symbols in the objective are outputs of preceding modules or ground-truth annotations: $\hat{\mathcal{S}}_i^{\text{complete}}$ is produced by the PoinTr reconstruction module, $\hat{s}_{qi,j}$ by the instance-mask head, $\hat{N}_q$ by the count head, $\hat{V}_{qi}$ and $\hat{\rho}_{qi}$ by the physical-estimation heads, and $\hat{w}_j^V$, $\hat{w}_j^\rho$, and $\hat{p}_j^{\text{comp}}$ by the physical-field predictor.

Inference Flow:

During inference, the calibrated RGBD point cloud $\mathcal{P}^{\text{RGBD}}$ and CSI tensor $\mathcal{Z}$ are first encoded into $\mathcal{G}^{\text{rgbd}}$ and $\mathcal{Q}$. The CSI path descriptors are back-projected into $\mathcal{R}$ and associated with RGBD points through $\mathbf{A}$. The associated CSI features are fused with RGBD features to produce $\mathbf{f}_j$, from which the weighted point cloud $\mathcal{P}^w$, physical fields, and instance graph are constructed. The resulting partial instances are projected into the PointLLM token space and processed using CSI-conditioned attention. The PointLLM representation is then passed to the PoinTr reconstruction module, which generates $\hat{\mathcal{S}}_i^{\text{complete}}$ and the refined instance $\tilde{\mathcal{S}}_i$. Finally, the physical-estimation heads produce $\hat{N}_q$, $\hat{\mathcal{S}}_{qi}$, $\hat{V}_{qi}$, and $\hat{\rho}_{qi}$. Hence, every output symbol is generated from the output of the preceding module, and no independent symbol is introduced outside the defined computational flow.

II. RESULTS AND EVALUATION

A. Evaluation Metrics

We evaluate the proposed framework using metrics that directly measure its three principal objectives: object counting, individual volume estimation, and individual density estimation. Point-cloud reconstruction quality is reported as a supporting metric because errors in recovered geometry can propagate to the final physical estimates.

Object-count metrics. Count MAE measures the mean absolute difference between the predicted and ground-truth fruit counts, whereas Count RMSE penalizes larger

counting errors more strongly. Count Accuracy denotes the percentage-based counting accuracy, and the Instance Matching/Recovery Rate (IMR) measures the proportion of ground-truth fruit instances correctly matched or recovered. Precision, Recall, and F-score evaluate instance-level fruit detection and reconstruction quality.

Volume-estimation metrics. Volume MAE measures the mean absolute error between predicted and ground-truth fruit volumes in cm^3 . Volume MAPE measures the mean absolute percentage error, while Volume Relative Error (RelErr) normalizes the volume error by the corresponding ground-truth volume. Volume Accuracy is derived from the volume relative error, with higher values indicating more accurate estimation.

Density-estimation metrics. Density MAE measures the mean absolute difference between predicted and ground-truth densities in g/cm^3 . Density MAPE reports the mean absolute percentage error, whereas Density Relative Error (RelErr) expresses the density error relative to the ground-truth density. Density Accuracy is computed from the density relative error, with higher values indicating better estimation.

3-D reconstruction metrics. Chamfer Distance (CD) measures the geometric discrepancy between the reconstructed and ground-truth point clouds, where lower values are better. The point-cloud F-score is the harmonic mean of precision and recall computed at a specified distance threshold, where higher values indicate more accurate reconstruction.

Occlusion and fusion metrics. The Error-Reduction Percentage measures the relative reduction achieved by RGBD-CSI fusion compared with RGBD-only processing. The Absolute Accuracy Gap denotes the difference in accuracy, in percentage points, between the fusion and RGBD-only configurations. Accuracy Drop measures the performance decrease as occlusion or the number of hidden objects increases, whereas Occlusion Robustness measures the degree to which performance is retained under increasing occlusion. Finally, the Recovery Rate denotes the percentage of hidden or occluded fruit instances successfully recovered.

The definition of each metric is given below:

Count Mean Absolute Error (Count MAE). Count MAE measures the average absolute difference between the predicted and ground-truth number of objects across all test piles:

$$\text{CountMAE} = \frac{1}{Q} \sum_{q=1}^Q |\hat{N}_q - N_q|, \quad (54)$$

where Q is the number of test piles, $\hat{N}_q$ is the predicted object count for pile q , and N_q is its ground-truth count; lower values indicate more accurate counting.

Count Accuracy. Count accuracy measures the fraction of test piles for which the predicted count exactly matches

the ground-truth count:

$$\text{CountAcc} = \frac{1}{Q} \sum_{q=1}^Q \mathbb{I}[\hat{N}_q = N_q], \quad (55)$$

where $\mathbb{I}[\cdot]$ is one when the condition is true and zero otherwise; higher values indicate more reliable pile-level counting.

Instance matching rate (IMR). Instance matching rate measures the proportion of ground-truth objects that are correctly matched to predicted instances using an assignment criterion such as the Hungarian algorithm and an IoU threshold δ_{IoU} :

$$\text{IMR} = \frac{1}{K} \sum_{i=1}^K \mathbb{I}[\text{IoU}(\hat{S}_i, S_i) \geq \delta_{\text{IoU}}], \quad (56)$$

where K is the number of ground-truth objects, $\hat{S}_i$ is a predicted instance, S_i is the matched ground-truth instance, and higher values indicate more accurate instance separation.

Volume mean absolute error (Volume MAE). Volume MAE measures the average absolute error between the predicted and ground-truth volumes of matched objects:

$$\text{VolumeMAE} = \frac{1}{K} \sum_{i=1}^K |\hat{V}_i - V_i|, \quad (57)$$

where $\hat{V}_i$ and V_i denote the predicted and ground-truth volumes of object i , respectively; lower values indicate more accurate volume estimation.

Volume relative error (RelErr). Volume relative error normalizes the volume error by the ground-truth volume, allowing comparison across objects of different sizes:

$$\text{VolumeRelErr} = \frac{100}{K} \sum_{i=1}^K \frac{|\hat{V}_i - V_i|}{V_i + \epsilon}, \quad (58)$$

where $\epsilon > 0$ prevents division by zero; lower values indicate better size-normalized volume estimation.

Density mean absolute error (Density MAE). Density MAE measures the average absolute difference between the estimated and ground-truth densities of matched objects:

$$\text{DensityMAE} = \frac{1}{K} \sum_{i=1}^K |\hat{\rho}_i - \rho_i|, \quad (59)$$

where $\hat{\rho}_i$ and ρ_i are the predicted and ground-truth densities of object i ; lower values indicate more accurate density estimation.

Density relative error. Density relative error expresses the density error as a percentage of the ground-truth density:

$$\text{DensityRelErr} = \frac{100}{K} \sum_{i=1}^K \frac{|\hat{\rho}_i - \rho_i|}{\rho_i + \epsilon}, \quad (60)$$

where $\epsilon > 0$ avoids numerical instability for densities close to zero; lower values indicate more reliable relative density estimation.

Completion F-score. The completion F-score com-

bines point-cloud precision and recall at a distance threshold δ and measures the proportion of geometrically correct reconstructed points:

$$F_\delta = \frac{2P_\delta R_\delta}{P_\delta + R_\delta}, \quad (61)$$

where P_δ is the fraction of predicted points within distance δ of the ground-truth point cloud and R_δ is the fraction of ground-truth points within distance δ of the prediction; higher values indicate better completion quality.

B. Ablation Study

We conduct a task-oriented ablation study to quantify the contribution of both the proposed ESP32 hardware/firmware design and the proposed Multi-Path PointLLM architecture. Since the primary objective is physical inventory under heavy occlusion, we focus on object counting and individual volume-density estimation rather than relying only on geometric completion metrics. We report count mean absolute error (Count MAE), exact count accuracy (Count Acc.), instance matching rate (IMR), volume mean absolute error (Volume MAE), volume relative error (Volume RelErr), density mean absolute error (Density MAE), and density relative error (Density RelErr). We also report reconstruction F-score as a supporting geometric metric. Lower values are better for error metrics, whereas higher values are better for accuracy, instance matching, and F-score.

The hardware ablations results in Table I show a progressive improvement as the wireless front-end becomes more informative. Replacing the complete ESP32 array with a single CSI link increases Count MAE from the illustrative full-model value of 0.54 to 1.19 and reduces instance matching from 82.4% to 59.6%, indicating that spatial diversity is important for distinguishing objects under heavy occlusion. Removing metadata coding further degrades performance because CSI measurements cannot be reliably assigned to their transmitting devices and time frames. Removing directional coding reduces the ability to distinguish path-dependent perturbations, while removing delay validation introduces physically inconsistent multipath components. These degradations also propagate to volume and density estimation because incorrect path association produces less reliable hidden-geometry evidence.

The PointLLM ablations show that geometry-aware fusion and multipath reasoning are important for the final physical-estimation task. Removing CSI back-projection increases Volume RelErr from the illustrative full-model value of 10.7% to 15.8%, demonstrating that CSI must be transferred into the RGBD coordinate system rather than appended as an unstructured global feature. Replacing gated fusion with direct concatenation causes additional count, volume, and density errors because the model cannot adaptively weight RGBD and CSI evidence. Removing the physical point fields has a particularly large effect on density estimation, increasing Density RelErr to 22.1%,

because local mass- and volume-related contributions are no longer available for instance-level integration. Removing CSI-conditioned pairwise attention reduces both instance matching and count accuracy, showing that relationships between multipath responses at different points contribute to object separation.

The largest geometric degradation is produced by removing the PoinTr reconstruction module. Without explicit completion, the model receives incomplete object geometry, reducing the illustrative F-score from 80.6% to 68.7% and increasing Volume RelErr from 10.7% to 20.9%. This result supports the role of reconstruction as an intermediate step for physical estimation: complete geometry is necessary for reliable integration of local volume weights, while CSI-conditioned reasoning improves the selection and refinement of the missing regions. Overall, the complete hardware-firmware and Multi-Path PointLLM configuration provides the best joint performance for object count, instance matching, individual volume, and individual density.

Interpretation. Count MAE and count accuracy evaluate whether the system discovers the correct number of objects, while IMR evaluates whether the discovered objects are correctly separated. Volume MAE and Volume RelErr measure the accuracy of individual geometric size estimates, whereas Density MAE and Density RelErr measure the accuracy of the final material-related estimates. CD and F_δ are supporting reconstruction metrics: they indicate whether the completed geometry is accurate enough to support reliable integration of local volume and density fields. Chamfer distance and F-score are commonly used complementary metrics for point-cloud completion evaluation [4], [5].

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Configuration	Count MAE↓	Count Acc.↑	IMR ↑	Volume MAE↓	Volume RelErr↓	Density MAE↓	Density RelErr↓	F-score ↑
RGBD only	1.42	61.0	54.8	18.7	24.9	0.31	28.6	61.8
ESP32 CSI, single link	1.19	66.7	59.6	16.4	22.1	0.27	25.1	64.9
ESP32 array, no metadata coding	0.98	72.1	65.3	14.1	19.5	0.23	21.7	68.8
ESP32 array, no directional coding	0.86	76.0	70.2	12.7	17.3	0.21	19.8	71.4
ESP32 array, no delay validation	0.79	78.4	72.0	11.9	16.1	0.20	18.7	72.5
Full hardware/firmware front-end	0.68	82.0	77.5	10.2	13.8	0.17	15.4	75.9
Full front-end, w/o CSI back-projection	0.74	79.7	74.1	11.6	15.8	0.19	17.6	73.8
Full front-end, concatenation fusion	0.72	80.4	75.0	11.2	15.1	0.18	16.8	74.6
Full front-end, w/o physical point fields	0.70	81.2	76.1	12.8	17.6	0.24	22.1	75.2
Full front-end, w/o CSI pairwise attention	0.75	79.3	73.8	11.4	15.5	0.20	18.1	74.1
Full front-end, w/o PoinTr reconstruction	0.83	76.8	69.5	15.3	20.9	0.25	21.9	68.7
Full Multi-Path PointLLM	0.54	86.7	82.4	8.1	10.7	0.13	11.8	80.6

TABLE I: Hypothetical ablation study covering both the ESP32 hardware/firmware novelty and the proposed Multi-Path PointLLM pipeline. Values are illustrative and must be verified experimentally. Count MAE, Volume MAE, Volume RelErr, Density MAE, and Density RelErr are lower-is-better; Count Acc., IMR, and F-score are higher-is-better.